

TBI Monte Carlo Calibration

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1 Measurements

For the following three geometries, we took measurements at the Cobalt unit and performed corresponding Monte Carlo simulations. The phantom for all measurements was solid water material (density 1.03 g/cm^3), dimensions $40 \times 40 \text{ cm}^2$ and 20 cm deep.

The ion chamber was at a depth of 10 cm on the central axis (gantry angle 0 degrees). The TG-51 calculation provided a reference to absolute dosimetry. The second and third geometries approximated treatment conditions during a TBI treatment. (Note: I have not used the second geometry data for anything at this time, but I included it here for completeness.)

1. TG-51 condition (SSD 80cm, depth 10cm, field size $10 \times 10 \text{ cm}^2$ at 80cm, 2 minutes)
2. Extended SSD with flattening filter (SSD 164.6 cm, depth 10cm, field size $35 \times 35 \text{ cm}^2$ at 80cm, 2 minutes)
3. Extended SSD with flattening filter, sweeping (same as above, swept 10 times $\rightarrow$ 4.5 minutes)

Readings:

1. TG-51 condition: (collimator angle 0 deg)

- -422.5 R
- -422.5 R
- -422.3 R

2. Extended SSD, no sweeps:

- -87.18 R (collimator angle: 0 deg)
- -87.20 R (collimator angle: 270 deg)
- -87.20 R (collimator angle: 270 deg)

3. Extended SSD, sweeping: (collimator angle 270 deg)

- -55.61 R
- -55.19 R
- -54.80 R

All readings were collected in Roentgens. The reading at the TG-51 condition was converted to absolute dose by accounting for source decay to the experiment date.

The first reading for the second geometry was performed without switching the collimator angle to 270 degrees. This was a mistake, although we figured it wouldn't make a huge difference, and we kept the measurement (and we have not used this data, anyway).

2 Calculations

2.1 Source decay

As of August 15 2017, the calculated TG-51 dose rate was 290.1 cGy/min at d_{max} . (half-life $T_{1/2} = 1921.5$ days, source decay constant $\lambda = 0.000360725$ days⁻¹).

The absolute calibration dated August 13, 2015, gave a dose rate of **373.7 cGy/minute** at TG-51 conditions (depth 0.5cm, the d_{max} for Co-60).

Our measurements were done on July 13 2017. The TG-51 dose rate at this time, by calculation from exponential decay law, was **290.3 cGy/min**.

$$\begin{aligned}\dot{D}(\text{July 13 2017}) &= \dot{D}(\text{Aug 13 2015}) \times \exp\left(-\frac{\ln(2) \times (700 \text{ days})}{1921.5 \text{ days}}\right) \\ \dot{D}(\text{July 13 2017}) &= 373.7 \text{ cGy/min} \times 0.7768 \\ \dot{D}(\text{July 13 2017}) &= 290.3 \text{ cGy/min}\end{aligned}$$

2.2 Conversion from MC Gy/particle to real dose

The MC-simulated dose at our measurement point for the TG-51 geometry was **2.3193×10^{-15} Gy/particle**. (I retrieved this value by using dosXYZshow on the cluster and checking in all dimensions to ensure I had the right point. I do not have a simple way of retrieving the error value at a given point from the .3ddose file, but looking over the file, the relative errors range from 0.7% to 3%. 3 billion histories used.)

We also need to scale the dose from our calculation above (section 2.1) down to our measurement point with a PDD. TG-51 is at d_{max} but our measurement

was at depth 10 cm. We do this with PDD data. (We do not try to measure at the top of the PDD curve because this is less forgiving to small positioning uncertainties).

$$PDD(d=10\text{cm}, r=10 \times 10 \text{ cm}^2, SSD=80\text{cm}, \text{Co}^{60}) = 0.564$$

(source: Ingrid's Phys535 tables)

This lets us calculate our dose rate $\dot{D}$ at our measurement point,

$$\begin{aligned}\dot{D}(\text{our measurement point}) &= \dot{D}(\text{TG-51}) \times PDD(\text{our measurement point}) \\ \dot{D}(10\text{cm}, 10 \times 10\text{cm}^2, 80\text{cm}) &= \dot{D}(d_{max}, 10 \times 10\text{cm}^2, 80\text{cm}) \times PDD(10\text{cm}, 10 \times 10\text{cm}^2, 80\text{cm}) \\ \dot{D}(10\text{cm}, 10 \times 10\text{cm}^2, 80\text{cm}) &= 290.3 \text{ cGy/min} \times (0.564) \\ \dot{D}(10\text{cm}, 10 \times 10\text{cm}^2, 80\text{cm}) &= \mathbf{163.73 \text{ cGy/min}}\end{aligned}$$

Our average ion chamber reading here was -422.433 R from 2 minutes. Our average ion chamber reading for our third geometry (extended SSD with flattening filter) was -55.20 R from 4.5 minutes (10 sweeps). So to get our dose rate at our third geometry measurement point,

$$\begin{aligned}\dot{D}(\text{third condition}) &= \dot{D}(10\text{cm}, 10 \times 10\text{cm}^2, 80\text{cm}) \times \frac{2 \text{ minutes} \times 55.20R}{4.5 \text{ minutes} \times 422.433R} \\ \dot{D}(\text{third condition}) &= \mathbf{9.51 \text{ cGy/min}}\end{aligned}$$

The MC-simulated dose at our measurement point for this third geometry was $\mathbf{1.190 \times 10^{-16} \text{ Gy/particle}}$.

We now compute the ratio (conversion factor) from Monte Carlo Gy/particle units to real Gy dose units. (We use Gy instead of cGy because our treatment planning systems require Gy inputs in the dose files, not cGy.) We are also going to divide by an exponential decay factor to see what our dose would have been when the source was calibrated - this means for future patients, we will just need to apply a scale factor depending on their treatment start date.

$$\begin{aligned}\text{general conversion factor} &= \frac{0.0951 \text{ Gy/min}}{1.190 \times 10^{-16} \text{ Gy/particle}} \div \exp\left(-\frac{\ln(2) \times (\text{days since } 13/08/15)}{1921.5 \text{ days}}\right) \\ \text{general conversion factor} &= 7.99160 \times 10^{14} \frac{\text{Gy/min}}{\text{Gy/particle}} \div \exp\left(-\frac{\ln(2) \times 700}{1921.5}\right) \\ \text{general conversion factor} &= 7.99160 \times 10^{14} \frac{\text{Gy/min}}{\text{Gy/particle}} \div 0.7768 \\ \text{general conversion factor} &= \mathbf{1.0288 \times 10^{15} \frac{\text{Gy/min}}{\text{Gy/particle}}}\end{aligned}$$

For each patient, we multiply this conversion factor by two patient-specific values:

- an exponential decay factor, $\exp(-\frac{\ln(2) \times (\text{days since 13/08/15})}{1921.5 \text{ days}})$
- six times the treatment time per field on the patient's treatment sheet (we end up with one prone simulation and one supine simulation, each one represents six treatment fields)

For example, for patient VC_TBI01 whose first treatment day was 25/01/16, 165 days after the source change on 13/08/15,

$$\text{conversion factor} = \text{general conv.fac.} \times \exp(\ln(2) \times \frac{165}{1921.5}) \times 6 \times \text{time per field}$$

$$\text{conversion factor} = 1.0288 \times 10^{15} \frac{\text{Gy/min}}{\text{Gy/particle}} \times (0.9422) \times 6 \times 6.88 \frac{\text{min}}{\text{field}}$$

$$\text{conversion factor} = 4.0014 \times 10^{16} \frac{\text{real Gy}}{\text{MC Gy/particle}}$$

This is the value that is currently coming up 20% short in the simulations.